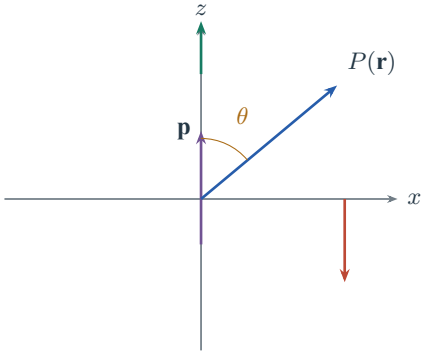
# M1-16 偶极子的场点、极角与两个检验方向




轴线电场与 **p** 同向；赤道电场与 **p** 反向

$$\varphi = \frac{\mathbf{p}\cdot\mathbf{r}}{4\pi\varepsilon_0 r^3}$$

$$\mathbf{E} = \frac{3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}}{4\pi\varepsilon_0 r^3}$$

$$\mathbf{E}_{\text{axis}} = \frac{2\mathbf{p}}{4\pi\varepsilon_0 r^3}$$

$$\mathbf{E}_{\text{eq}} = -\frac{\mathbf{p}}{4\pi\varepsilon_0 r^3}$$